David Andrews

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Education

Georgia Institute of Technology, Atlanta, GA: *B.S. in Computer Science* Aug. 2023 – May 2026
GPA: 4.0/4.0
Courses: DL, ML, Systems for ML, Intro to AI, Perception and Robotics, Algorithms, OS

Experience

Georgia Tech, Chao Zhang Lab: *Undergraduate Researcher, Artificial Intelligence* Aug. 2024 – Present
- Built PRM-guided tree-search reasoning systems with high-throughput inference using SGLang and vLLM caching to generate synthetic trajectories with error correction and exploration.
- Trained multi-turn LLM agents with GRPO on SWE-bench, SWE-Smith, and SWE-Gym, adding reward-hacking defenses, tool-use checks, and rollout instrumentation.
- Generalized the system to Docker/MCP tool environments and scaled asynchronous PPO/GRPO/RLVR training using AReaL on Ray with a Kubernetes-managed sandbox server.

Georgia Tech, Qirun Zhang Lab: *Undergraduate Researcher, Systems* Jan. 2026 – Present
- Developing an LLM agent harness for automated bug finding and fixing in the LLVM compiler, with specialized tool skills and instructions for navigating large-scale systems codebases.
- Investigating agent-based approaches to accelerate commit bisection, reducing the manual effort of identifying bug-introducing changes in complex compiler histories.

Georgia Tech, EIC Lab: *Undergraduate Researcher, Hardware Design* Feb. 2026 – Present
- Curating a SWE-bench-style benchmark for Verilog/RTL hardware design, derived from real-world GitHub PRs and evaluated using Verilator, Yosys, and Icarus Verilog for functional verification.
- Developing an agent harness to benchmark frontier models on industry-relevant hardware design tasks and exceed existing agent performance on the benchmark.

DuckAI: *AI Researcher & Engineer* Aug. 2023 – Present
- Engineered **DuckTrack**, a cross-platform computer-use capture tool that records screen, keyboard, and mouse with sub-frame alignment via OS input APIs and OBS synchronization.
- Enabled high-quality computer-use datasets that were later adopted in OSWorld and OpenCUA for benchmarking and foundation model training.
- Designed and implemented **PRMBench**'s full-stack annotation platform (React, FastAPI, SQLite), prototyping UI workflows with annotators to improve labeling speed and inter-annotator agreement for process reward model datasets.
- Built large-scale TTS data pipelines (YouTube and podcast ingestion, upscaling, VAD, diarization) and contributed optimizations to insanely-fast-whisper to accelerate transcription for speech datasets.

Selected Projects

Tool-Integrated RL Agent Training System May 2025 – Present
- Engineered a general tool-using agent framework with Docker sandboxes and an MCP interface (browser, terminal, code execution), supporting multi-turn rollouts across math, search, SWE-bench, and puzzle tasks.
- Integrated AReaL for asynchronous PPO/GRPO/RLVR training to maximize throughput under high tool latency, coordinating rollout workers with sandboxes hosted on a Kubernetes cluster.

DuckTrack Nov. 2023 – Jan. 2024
- Extended and maintained precise, cross-platform capture of screen, keyboard, and mouse events by benchmarking native input APIs and synchronizing with OBS to achieve millisecond-level, sub-frame alignment for computer-use datasets.

Emergent Misalignment Oct. 2025 – Jan. 2026
- Replicated and investigated emergent misalignment arising from benign fine-tuning of LLMs; findings published on LessWrong.

AlphaEvolve Extensions May 2025 – Sept. 2025
- Replicated circle-packing results and extended OpenEvolve with new combinatorics and compiler tasks, upstreaming improvements and exploring sphere-packing and other scientific application domains.

Skills

Languages: Python, Rust, C, C++, Go, JavaScript, Bash, Verilog
ML: PyTorch, vLLM, SGLang, Ray, AReaL, veRL, HuggingFace
Systems: Linux, Docker, Kubernetes, SLURM, AWS, FastAPI, SkyPilot, uv, Git

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